

Master's thesis / Bachelor's thesis

Robotic Systems in Battery Manufacturing: Potentials, Applications, and Implementation Strategies

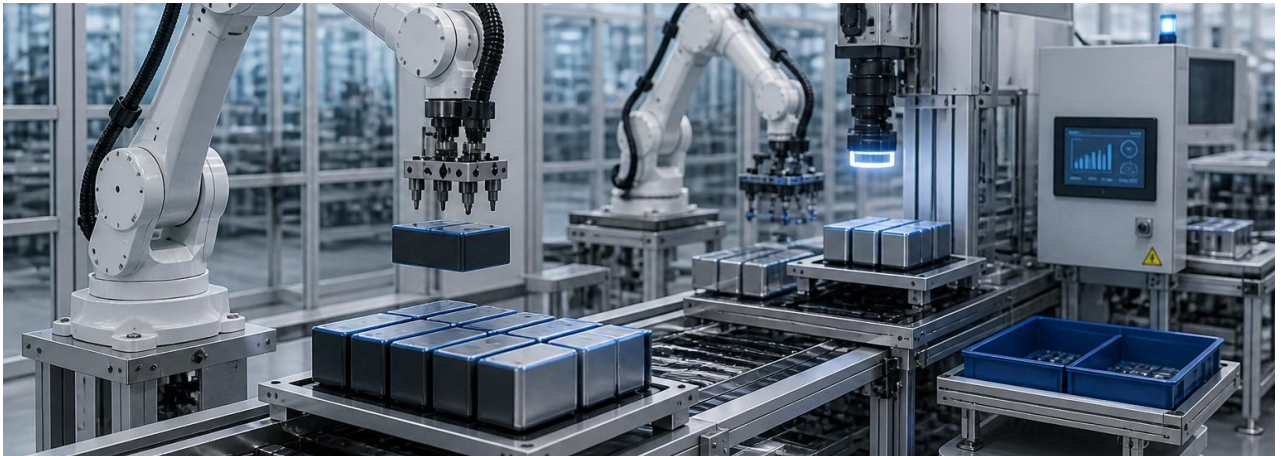


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Initial situation:

The production of lithium-ion batteries is rapidly scaling worldwide, driven by demand for electric mobility and energy storage. New large-scale manufacturing facilities, particularly in Europe, must meet high requirements in terms of quality, efficiency, and cost. At the same time, battery production is characterized by complex processes, sensitive materials, and strict environmental conditions such as dry room operation. Despite increasing automation, many process steps remain partially manual, especially in cell assembly, material handling, logistics, and quality assurance. These manual interventions limit scalability and introduce variability. Robotic systems offer significant potential to improve process stability, increase efficiency, enhance safety, and enable better data integration for digitalization. However, challenges remain regarding integration, handling of delicate materials, operation under dry room conditions, and economic feasibility.

Your task:

The objective of this work is to analyze and evaluate the use of robotic systems along the battery cell production value chain. The study begins with an overview of current manufacturing processes and their specific requirements. Based on this, suitable application areas for robotics are identified, including industrial robots, collaborative robots, and mobile systems. Typical use cases include material handling, automated inspection, flexible assembly, and intralogistics. These applications are then assessed from a technical and economic perspective, considering factors such as automation potential, cost, quality impact, and scalability. Building on this, selected use cases are developed into concrete application scenarios, taking into account different cell formats and production environments. Finally, the work outlines future trends in robotics and provides recommendations for industrial implementation and further research.

Prerequisites:

- Degree in (industrial) engineering (or comparable)
- Enthusiasm for battery technologies
- Commitment and willingness to learn

We offer:

- Insight into the battery technology of today and tomorrow
- Electromobility laboratory (eLab) at RWTH Aachen University
- Option to shape the thematic focus

Interested?

Please send a current transcript of records as well as your CV and certificates to the e-mail address below.

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